

COMPONENTS

Diode

Vishay Intertechnology has introduced 28 new 600V and 650V FRED Pt gen 4 ultra-fast recovery diodes optimised for high-frequency converters in power modules, motor drives, UPSes, solar inverters and welding machine inverters.

Offered as die in wafer form, the devices offer ultra-low forward voltage and reverse recovery charge to reduce losses and increase efficiency. Also, their extremely soft turn-off behaviour minimises over voltages under all switching conditions.

The devices are designed for use as anti-parallel diodes in combination with Vishay's Trench insulated gate bipolar transistors (IGBTs).

Vishay Intertechnology Inc.

Website: www.vishay.com

MOSFET

Toshiba has launched SSM6J507NU, a P-channel MOSFET, suitable for load switches of USB type C terminals conforming to the USB PD specifications. USB PD is a technology that allows a power supply of up to 100W (20V/5A), while a load switch between a USB type C terminal and battery charger integrated circuit (IC) is required to have an isolation voltage of 20V or higher.

The MOSFET is suitable for this application as it achieves $VD_{SS} = -30V$ and a low ON resistance.

Toshiba India Pvt Ltd

Website: www.toshiba-india.com

ADC

Texas Instruments (TI) has announced ADS54J60, a 16-bit 1GSPS analogue-to-digital converter.

Key features of ADS54J60 are:

1. Allows higher frequencies with more accurate signal analysis
2. Offers more than 3dB better

SNR (70dBFS at $F_{IN} = 170MHz$), low noise floor of $-159dBFS/Hz$ and spurious-free dynamic range (SFDR) of 86dBc

3. Offers two channels in a 10mmx10mm package

4. Reduces data interface speed and external processor resources

5. Offers 20 per cent low power consumption

Texas Instruments

Website: www.ti.com

DSP chip

ON Semiconductor has announced the release of BelaSigna 300 AM with AfterMaster HD Audio Labs Inc. BelaSigna 300 AM digital signal processing chip embedded with AfterMaster technology is an audio solution that enhances the listening experience on any consumer device.

ON Semiconductor

Website: www.onsemi.com

Controller

The STNRG digital-controller family from STMicroelectronics helps designers maximise the advantages



of digital power conversion, including high efficiency under all load conditions, enhanced safety, rich diagnostics and convenient network connectivity.

STNRG ICs contain ST's state machine event driven (SMED) high-resolution pulse-width modulation (PWM) generator, in combination with an STM8 based supervisory core. The devices integrate 32kB EEPROM, 6kB RAM, an ADC, op-amp, I2C port and GPIOs.

STMicroelectronics

Website: www.st.com

Intelligent power module

Infineon Technologies has launched an intelligent power module (IPM), MIPAQ Pro, which provides an all-in-one solution for a wide spectrum of scalable and compact inverter designs to be implemented in wind, solar and industrial drives applications. It is a fully qualified and tested IPM, integrating IGBTs, gate drivers, a heat sink, sensors, digital control electronics and digital bus communication in one robust device.

Infineon Technologies AG

Website: www.infineon.com

Front-end power module

Vicor has launched high-density, low-profile, integrated VIA PFM AC-DC front-end power modules that are capable of achieving a power density of $8W/cm^3$ ($127W/in^3$), supplying an isolated, PFC-regulated 24V or 48V SELV DC output at up to 400W from the universal AC input range of 85V to 264V with 93 per cent peak efficiency.

Vicor Corp.

Website: www.vicorpower.com

TVS diode array

Littelfuse has introduced the miniature four-channel bi-directional SPI015 series TVS diode array (SPA diodes), designed to protect data lines that may experience destructive electrostatic discharge (ESD). These robust diodes can safely absorb repetitive ESD strikes ($\pm 20kV$ contact, $\pm 30kV$ air) well above the maximum level specified in the IEC61000-4-2 international standard without performance degradation.

Littelfuse Inc.

Website: www.littelfuse.com

Companion chip

The MB86R91 APIX companion chip enables the connection of modern high-performance ap-